

NATS Migration Strategy — one Australian cluster to a leafnode hub with two leaves

A revised, six-phase path from a single pilot cluster in Australia to Demo 03's T5 shape: a control hub in Africa with an Australian leaf and a South African leaf. Every phase has a gate. Cluster = 3 NATS instances throughout.

What changed from the first draft

- **The domain goes in on day one.** The original plan converted a live, data-holding Australian cluster to a leaf with a `jetstream.domain` in phase 3. That restarts every server and changes the estate-wide JetStream API address from `$JS.API` to `$JS.au.API`. Set on an empty cluster, it costs nothing. The conversion phase disappears.
- **Moving the hub and adding a region are two phases.** The original phase 4 did both at once.
- **Every phase has a gate.** A named test must pass before the next phase starts.

Phase 0 — names and subjects, on paper

PLACE	DOMAIN	CLUSTER NAME	SERVERS
Hub	hub	hub-cl	hub-1 / hub-2 / hub-3
AU	au	au-cl	au-1 / au-2 / au-3
ZA	za	za-cl	za-1 / za-2 / za-3

- **Server names are permanent.** Every `server_name` is globally unique and never changes afterwards.
Renaming in a live JetStream cluster breaks it.
- **Cluster names are distinct.** Every `cluster.name` differs. A leaf cluster must not share the hub’s cluster name or the server rejects the link.

SUBJECT	AUTHORITATIVE PUBLISHER	OWNING DOMAIN	COPIED WHERE, AND HOW	IDEMPOTENCY KEY
Example: ...				
Example: ...				
Example: ...				

The matrix is the only prevention for silent double capture. T5 can store a deliberate copy in two places; the written ownership rule is what makes that copy intentional, per `D03-R6` . **Accounts are the tenant boundary.** They are not the domain or the cluster (`D03-R2`). Cross-account sharing is explicit export/import with subject renaming (`D03-R8` , variant E).

Gate 0. The matrix is written and signed off. No servers exist yet.

PHASE 1 One Australian cluster, built leaf-ready

AUSTRALIA



— solid arrow = client request

■ cyan boundary = messaging cluster

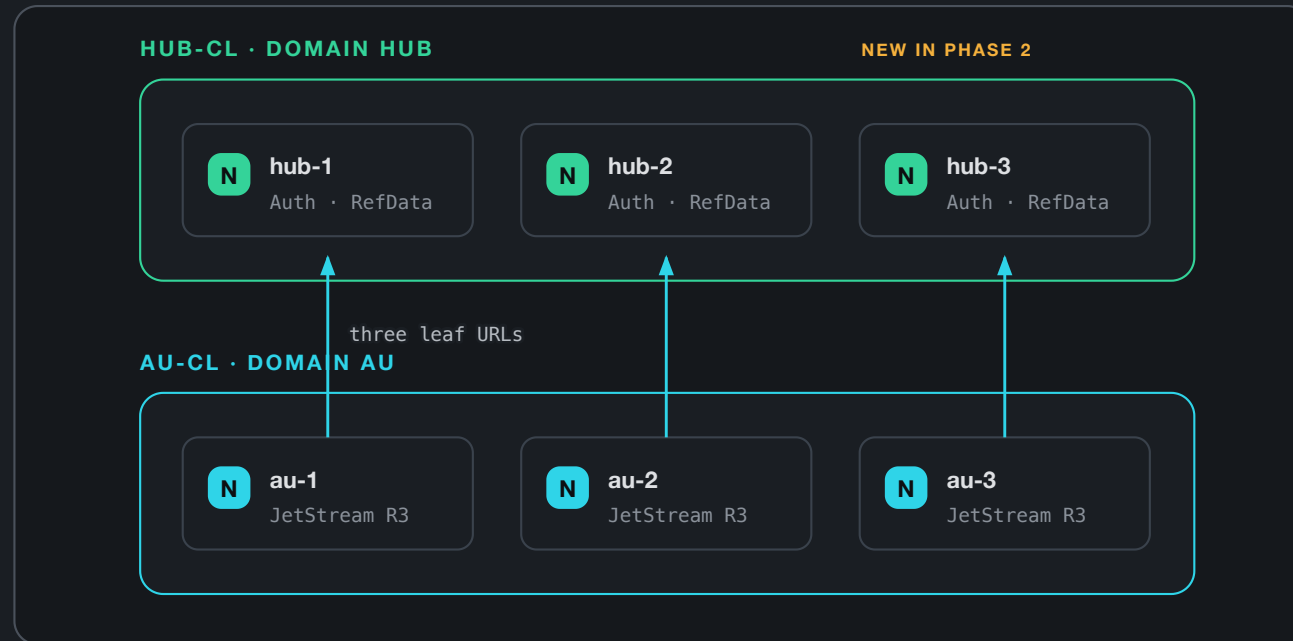
Set the irreversible names before data exists. The client uses `$JS.au.API` from day one, so its API address never changes. DEMO 03 · T5 · D03-R1

Domain, cluster name, server names and accounts are set at first boot. Clients are domain-aware from day one, so the API address never changes again.

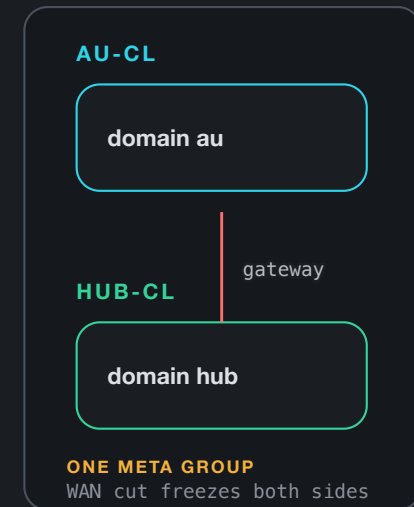
Gate 1. The pilot runs; streams and KV buckets work; every client is domain-aware.

PHASE 2 Add the control hub, join it by leaf link

AUSTRALIA



REJECTED



solid arrow = leaf link

red = rejected failure shape

cyan = messaging · green = shared control plane

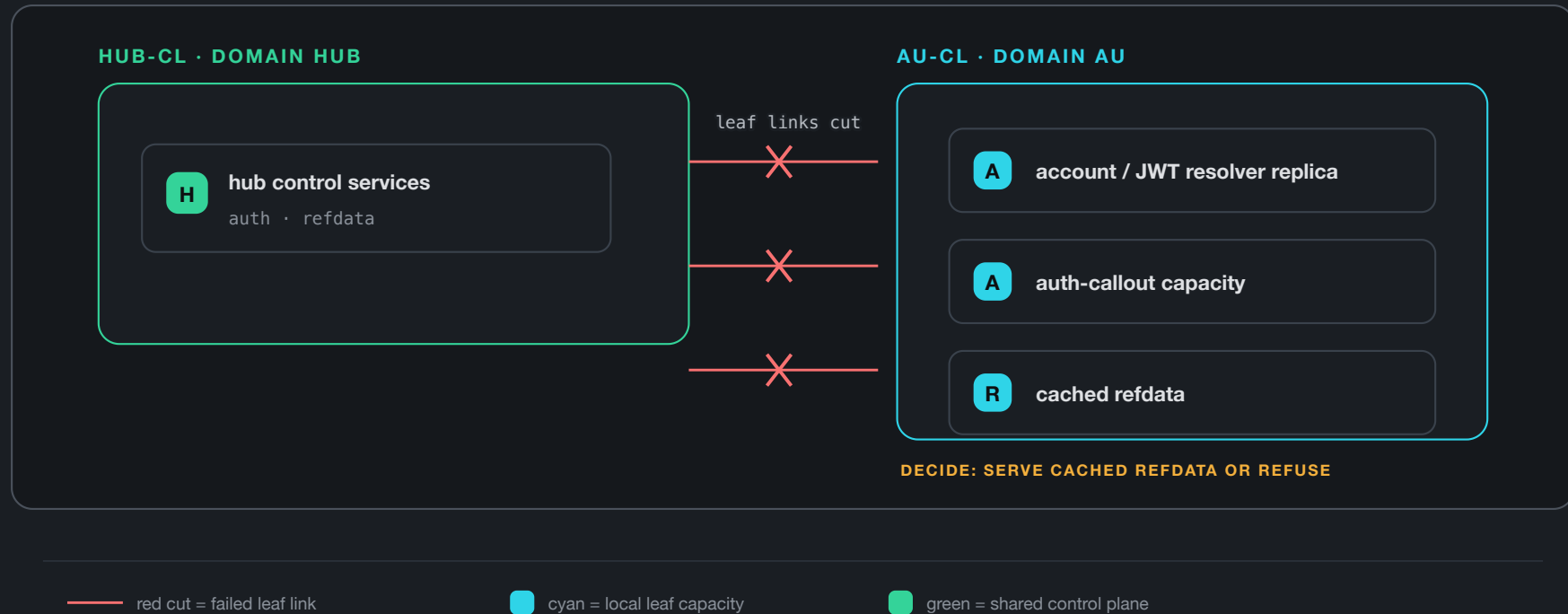
Leaf links preserve three independent JetStream meta groups. Every leaf server lists all three hub leaf URLs. A gateway is rejected: one meta group makes a WAN cut freeze both sides (**D03-R3** , checks **B3 / B6**). **DEMO 03 · T5 · D1**

This is already T5 with one leaf, so the original phase 3 has disappeared. Use a leaf link, not a gateway. Put all three hub URLs on every leaf server; one URL makes one hub node a single point of failure.

Gate 2. (1) Stop the whole hub. AU still elects a leader, stores publishes and creates a new 3-replica stream (**D10-D14**). (2) Every cross-domain mirror carries `external: { api: "$JS.hub.API" }` . A missing prefix gives a healthy-looking stream with zero messages and no error (**D03-R7**); alert on mirror lag, not mirror existence. (3) The subject matrix matches reality. Nothing is stored twice.

PHASE 3 Make auth survive the hub being gone

AUSTRALIA · FAILURE TEST



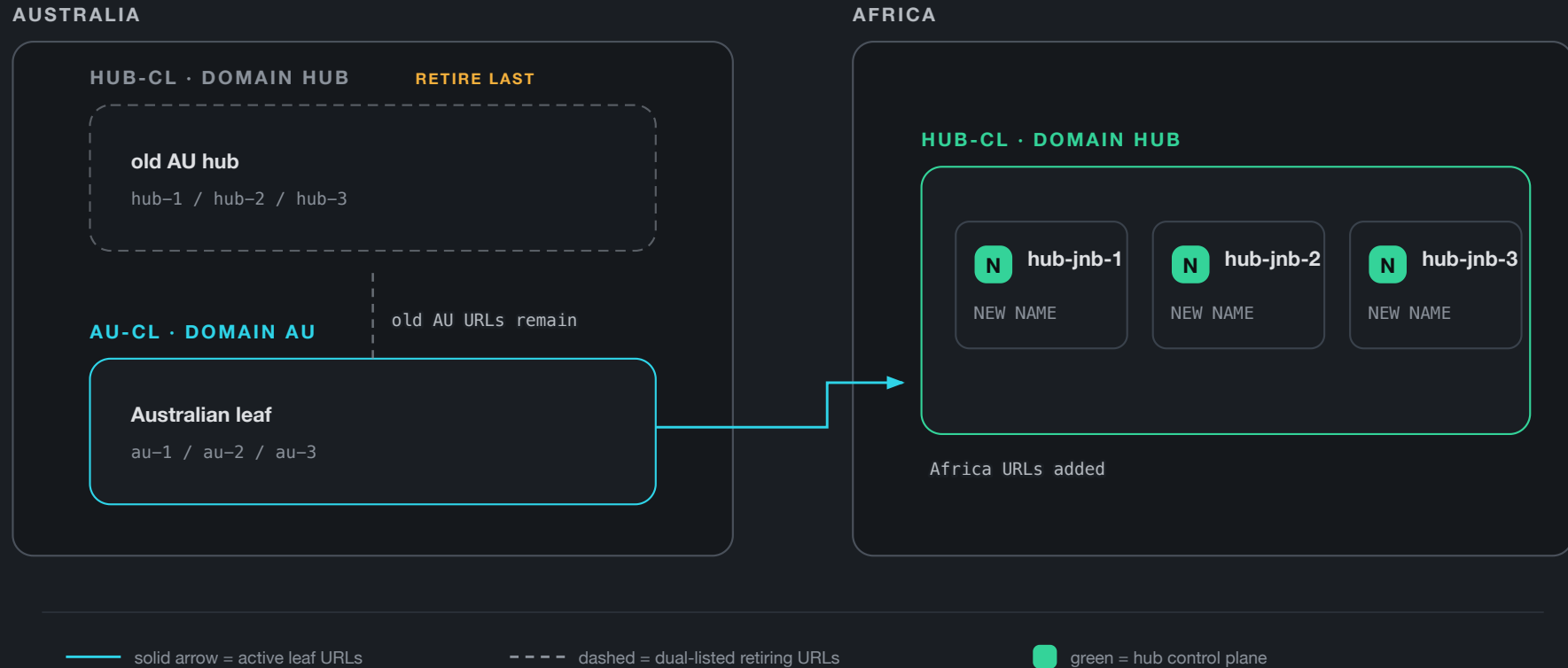
Authentication must have a local answer when the hub is absent. This figure makes the disconnected AU decision explicit. Demo 03 measured hub loss, not operator-mode JWT behaviour. [DEMO 03](#) · [T5](#) · [D10-D14](#)

Do this while the hub is still next door and a wrong answer is free. Answer three questions separately: (1) account/JWT lookup: put a resolver copy in AU; (2) auth callout / token issuer: put capacity in AU or AU cannot issue new tokens when the link drops; (3) leaf credentials: decide where they live and who rotates them.

Demo 03 used plain user/password auth in hand-written `.conf` files. It offers **no evidence** for operator-mode JWT behaviour. This phase is unmeasured and must be tested, not assumed. Decide and write down whether AU serves cached refdata or refuses when the hub is gone.

Gate 3. Cut the hub link for 30 minutes. AU keeps authenticating, keeps serving refdata, and nothing queues forever.

PHASE 4 Move the hub to Africa. The hub only.



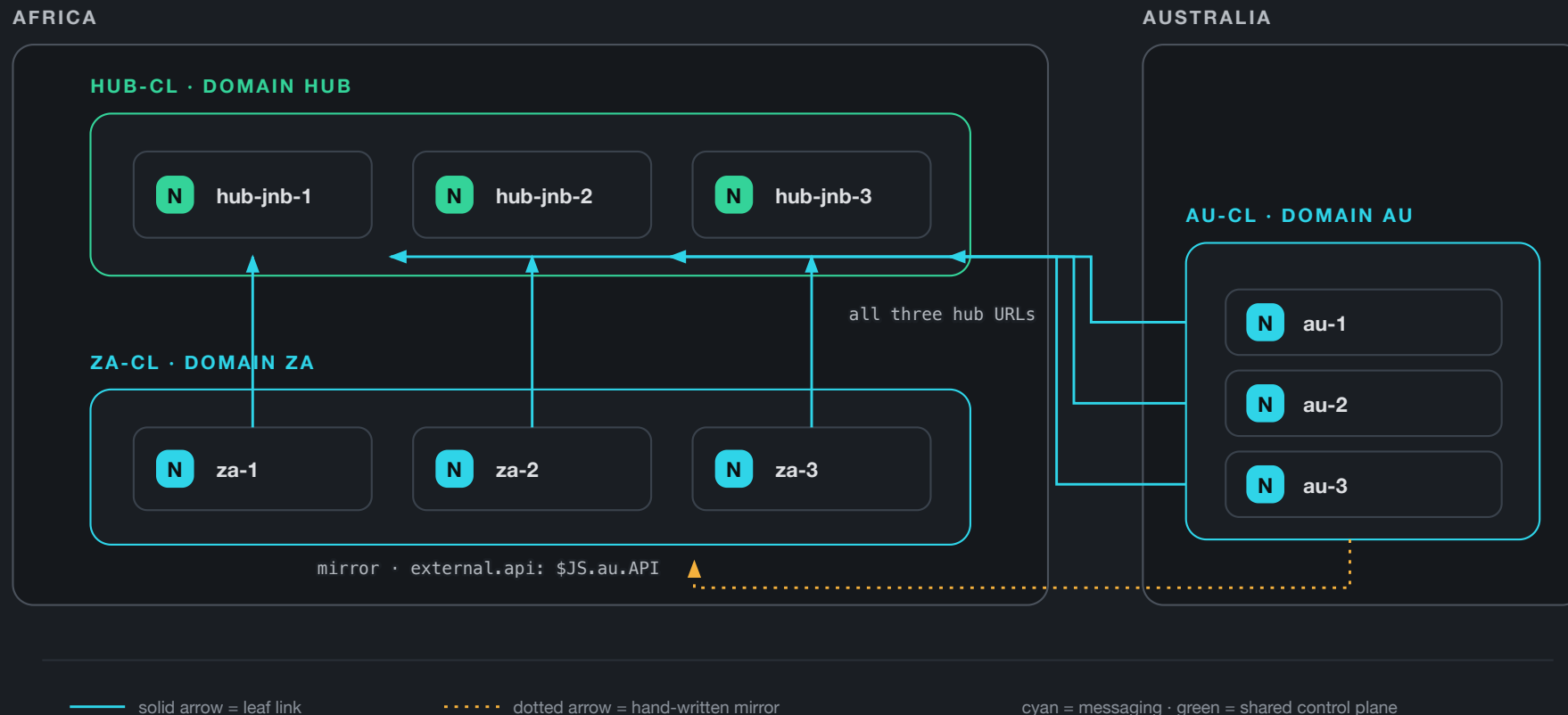
Move only the hub, with both URL sets listed during the cutover window. The Africa hub keeps domain and cluster name, but every server name is new. [DEMO 03](#) · [T5](#) · [D03-R1](#)

1. Build the Africa hub with the same domain and cluster name but **new server names**; never reuse a name.
2. Copy hub state across: auth and refdata only, so small.
3. Add the three Africa leaf URLs alongside the Australian ones.
4. Verify.
5. Soak for a week.
6. Only then remove the Australian URLs and shut the old hub down.

AU now pays ocean latency to reach auth. Phase 3 is what makes that survivable.

Gate 4. AU runs on the Africa hub for a full week. Repeat the Gate 2 hub-loss test.

PHASE 5 Add the South African leaf. Full T5.



Full T5 has three independent three-node JetStream clusters. Nothing copies itself across a leaf link. Every cross-region copy is a hand-written mirror with the other domain's external API. DEMO 03 · T5 · D1, D03-R6, D03-R7

ZA points straight at its final hub and never does a URL swap. Every AU↔ZA copy is a mirror written by hand with `external.api` pointing at the other domain, because nothing copies itself across a leaf link.

Gate 5. Run full T5. Kill the hub and both AU and ZA keep working (D10-D14). Publish to an AU-owned subject and confirm it is stored once, not twice.

Still unmeasured — do not assume these

- **The hub as a real store.** Demo 03’s T5 hub only relayed. This plan’s hub holds auth and refdata. If that means JetStream streams on the hub, it is untested here; Demo 03’s own “Still open” list says so.
- **Mirror catch-up at production volume.** No lag or bandwidth figure exists in either shape. Measure it in Phase 2.
- **Leaf reconnect with many regions.** Only two leaves were ever measured.
- **Operator-mode auth across a leaf link.** Demo 03 used plain passwords. Phase 3 is where this is measured for the first time.

Old plan versus new

ORIGINAL FOUR PHASES	REVISED SIX PHASES
Phase 1: AU cluster	Phase 0: names + subject matrix on paper
Phase 2: + hub	Phase 1: AU cluster, domain <code>au</code> from day one
Phase 3: convert AU to a leaf	Phase 2: + hub, leaf link (old 2 and 3 merged; old 3 disappears)
Phase 4: move hub + add ZA	Phase 3: make auth survive hub loss Phase 4: move hub to Africa, alone Phase 5: add ZA leaf